

Jae-Won Chung

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Summary

I am a fifth year PhD candidate in CSE at the University of Michigan, advised by [Professor Mosharaf Chowdhury](#). I build **efficient software systems for machine learning**, with a focus on the efficient management of not only time, but also **energy**. My research views energy as a first-class systems resource that is worth carefully optimizing and allocating based on precise measurement and understanding.

I am passionate about **open-source** software and making real-world impact with my research. My open-source works, including the [Zeus](#) library, have received wide recognition from academia and industry from for instance NVIDIA, [Google](#), Microsoft, [PyTorch Foundation](#), and [GitHub](#). I created and lead the [ML.ENERGY initiative](#) as part of my research and open-source efforts, which is now a cross-institutional effort.

Education

University of Michigan

PH.D. CANDIDATE IN COMPUTER SCIENCE AND ENGINEERING

Ann Arbor, MI, USA

Sep 2021 - present

University of Michigan

M.S. IN COMPUTER SCIENCE AND ENGINEERING

Ann Arbor, MI, USA

Sep 2021 - Apr 2023

Seoul National University

B.S. IN ELECTRICAL AND COMPUTER ENGINEERING

Seoul, South Korea

Mar 2015 - Aug 2021

- GPA: 4.04/4.3 (overall) 4.15/4.3 (major), Summa Cum Laude. Period includes two years of military service.

Publications

Peer-reviewed conference publications

* Equal contribution

- **GPU-to-Grid: Voltage Regulation via GPU Utilization Control**, Zhirui Liang, [Jae-Won Chung](#), Mosharaf Chowdhury, Jiasi Chen, Vladimir Dvorkin, **PowerUp**, 2026
- **The ML.ENERGY Benchmark: Toward Automated Inference Energy Measurement and Optimization**, [Jae-Won Chung](#), Jeff J. Ma, Ruofan Wu, Jiachen Liu, Oh Jun Kwon, Yuxuan Xia, Zhiyu Wu, Mosharaf Chowdhury, **NeurIPS Datasets & Benchmarks track (spotlight)**, 2025 (Spotlight acceptance rate = 2.81%)
- **Reducing Energy Bloat in Large Model Training**, [Jae-Won Chung](#), Yile Gu, Insu Jang, Luoxi Meng, Nikhil Bansal, Mosharaf Chowdhury, **SOSP**, 2024 (Acceptance rate = 17.34%)
- **Zeus: Understanding and Optimizing GPU Energy Consumption of DNN Training**, Jie You*, [Jae-Won Chung*](#), Mosharaf Chowdhury, **NSDI**, 2023 (Acceptance rate = 18.38%)
- **ShadowTutor: Distributed Partial Distillation for Mobile Video DNN Inference**, [Jae-Won Chung](#), Jae-Yun Kim, Soo-Mook Moon, International Conference on Parallel Processing (**ICPP**), 2020 (Acceptance rate = 28.99%)

Preprints, technical reports, and workshop publications

- **OpenG2G: A Simulation Platform for AI Datacenter-Grid Runtime Coordination**, [Jae-Won Chung*](#), Zhirui Liang*, Yanyong Mao, Jiasi Chen, Mosharaf Chowdhury, Vladimir Dvorkin, 2026
- **Cornserve: A Distributed Serving System for Any-to-Any Multimodal Models**, [Jae-Won Chung*](#), Jeff J. Ma*, Jisang Ahn, Yizhuo Liang, Akshay Jajoo, Myungjin Lee, Mosharaf Chowdhury, **ACM CAIS Demos**, 2026
- **Where Do the Joules Go? Diagnosing Inference Energy Consumption**, [Jae-Won Chung](#), Ruofan Wu, Jeff J. Ma, Mosharaf Chowdhury, 2026
- **Kareus: Joint Reduction of Dynamic and Static Energy in Large Model Training**, Ruofan Wu, [Jae-Won Chung](#), Mosharaf Chowdhury, 2026
- **Cornserve: Efficiently Serving Any-to-Any Multimodal Models**, Jeff J. Ma*, [Jae-Won Chung*](#), Akshay Jajoo, Myungjin Lee, Mosharaf Chowdhury, 2025
- **Toward Cross-Layer Energy Optimizations in AI Systems**, [Jae-Won Chung](#), Nishil Talati, Mosharaf Chowdhury, **DOE ASCR Energy-Efficient Computing for Science Workshop**, 2024

- **Andes: Defining and Enhancing Quality-of-Experience in LLM-Based Text Streaming Services**, Jiachen Liu, [Jae-Won Chung](#), Zhiyu Wu, Fan Lai, Myungjin Lee, Mosharaf Chowdhury, 2024
- **Chasing Low-Carbon Electricity for Practical and Sustainable DNN Training**, Zhenning Yang, Luoxi Meng, [Jae-Won Chung](#), Mosharaf Chowdhury, **ICLR Workshop: Tackling Climate Change with Machine Learning**, 2023

Open Source Projects

- **Zeus** (☆356 ♪ 45), [A framework for deep learning energy measurement and optimization](#). **A PyTorch ecosystem project. GitHub secure open source fund cohort**. Provides a unified abstraction over various CPU, GPU, and SoC vendors, and an array of time & energy optimizers.
- **Cornserve** (☆124 ♪ 10), [An efficient serving platform for any-to-any multimodal models](#). Takes a microservice approach to serving complex-structured multimodal input & output models.
- **OpenG2G** (☆9 ♪ 0), [Modular simulation library for AI datacenter-grid interaction](#). Enables research on how AI datacenters interact with the power grid through composable, extensible simulation modules.
- **BERT4Rec-VAE-Pytorch** (☆415 ♪ 99), [A PyTorch framework for recommendation model training](#), with abstract classes for pluggable model, dataset, and samplers. BERT4Rec and Netflix VAE models implemented.
- **Reason** (☆197 ♪ 4), [A shell for managing research papers, written in Rust](#). Supports importing papers from file and URL, attaching markdown notes, and creating an HTML book with notes. Uses `serde` to persist data in human-readable and cloud sync-friendly format.
- **Pegasus** (☆33 ♪ 3), [An SSH command runner with a focus on simplicity, written in Rust](#). Runs multiple commands asynchronously using the `tokio` runtime and streams stdout and stderr back to the user. Battle-tested through multiple research projects and benchmarking.

Number of stars and forks are up-to-date as of May 7th, 2026.

Honors & Awards

Apr 2026	MLSys Rising Stars , MLCommons Machine Learning and Systems Rising Stars 2026 cohort	MLCommons
Nov 2025	Slingshot Grant , \$25,000 USD AWS Credits for the development of the ML.ENERGY Initiative	Laude Institute
Aug 2025	GitHub Secure Open Source Fund , \$10,000 for the development of the Zeus project	GitHub
May 2024	PyTorch Ecosystem Project , Zeus was included in the PyTorch Ecosystem	PyTorch Foundation
Jan 2024	Research award , \$20,000 for the development of the ML.ENERGY Initiative	Salesforce
Jan 2024	Mozilla Technology Fund 2024 , \$50,000 for the development of the Zeus project	Mozilla
Nov 2022	Carbon Hack '22 Second Best Solution , Carbon-Aware DNN Training with Zeus , \$25,000	Green Software Foundation
Jul 2021	Kwanjeong Overseas Scholarship , \$25,000	Kwanjeong Educational Foundation
Mar 2019	Kwanjeong Undergraduate Scholarship , \$20,000 over two years	Kwanjeong Educational Foundation

Talks

Apr 2026	Energy-Efficient Machine Learning Systems	UC Berkeley eMERGE Seminar
Dec 2025	Energy and Power as First-Class ML Design Metrics (w/ NVIDIA, Google, Meta)	NeurIPS 25 Tutorial
Oct 2025	Energy as a First-Class ML Design Metric	UW-Madison MadSystems Seminar
Sep 2025	Power and Energy as First-Class AI Design Metrics	KPAI (Bay Area Korean AI Meetup)
Jun 2025	Energy as a First-Class Resource in Machine Learning Systems	Pruna AI
May 2025	Energy-Efficient Systems for Machine Learning	Harvard Power and AI Initiative
Nov 2024	Energy-Efficient Systems for Machine Learning	SOSP 24 Doctoral Workshop
Apr 2024	Power and Energy Considerations in Machine Learning Systems	University of Michigan (EECS 598)
Oct 2023	Energy-Efficient Software Systems for Machine Learning	Seoul National University
Oct 2023	Energy-Efficient Deep Learning with PyTorch and Zeus	PyTorch Conference
Sep 2023	Energy-Efficient Deep Learning with Zeus	Massachusetts Institute of Technology

Selected Media Coverage

My research and open-source works have been covered by various media outlets, including MIT Technology Review, ArsTechnica, and Science News.

Jul 2025	<u>How much energy does your AI prompt use? It depends.</u>	<i>Science News</i>
May 2025	<u>We did the math on AI's energy footprint. Here's the story you haven't heard.</u>	<i>MIT Technology Review</i>
May 2025	<u>AI Consumes Lots of Energy. Can It Ever Be Sustainable?</u>	<i>The New Stack</i>
Mar 2025	<u>Can we make AI less power-hungry? These researchers are working on it.</u>	<i>ArsTechnica (front page)</i>
Nov 2024	<u>Up to 30% of the power used to train AI is wasted: A software tool could help fix that.</u>	<i>Tech Xplore</i>
Apr 2023	<u>University of Michigan's 'Zeus' Framework Downsizes AI's Massive Carbon Footprint.</u>	<i>HPCWire</i>
Apr 2023	<u>Researchers claim they can cut AI training energy demands by 75%.</u>	<i>DatacenterDynamics</i>

Service

- **Ad-hoc reviewer**, Nature
- **Systems Reading Group Organizer**, Paper reading group inside Michigan CSE. Organizer from September 2022 to December 2025.

Teaching

- **Systems for Generative AI (UMich, Fall 25)**, GSI, three lectures on GenAI and GenAI systems fundamentals. **Instructor evaluation median 5.00 (highest possible)**. Written comments included phrases “excellent” (three times), “perfect,” and “one of the best.”
- **Operating Systems (SNU, Spring 21)**, Lead TA, managed Linux kernel hacking projects and led student team design reviews.
- **Computer Architecture (SNU, Fall 20)**, Peer tutor, provided 30 hours of online lecture. **Best Tutor Award**.